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ABSTRACT

Recent work modeling the rheological behavior of human blood indicates that blood has all the hallmark features of a complex material, including shear-thinning, viscoelastic behavior, yield stress, and thixotropy. Using a recently developed linear superposition technique to account for the effects of thixotropy with the Giesekus model and recently collected human blood rheological data from a strain-controlled rheometer, we perform parametric and statistical analysis of the parameter values of 5 donors. The work is validated with the incorporation of a recent thixotropic framework to model elastic and viscoelastic contributions from the microstructure. The elastic and viscoelastic stress contributions from the microstructure are then linearly superimposed with the viscoelastic backbone solution for stress given by the classic Giesekus rheological model. Demonstrated here are a parametric and statistical analysis and a comparison of the ability of the new enhanced thixotropic Giesekus model to predict large amplitude oscillatory shear and uni-directional large amplitude oscillatory shear flow. In addition, there is a new methodology to model the normal forces of blood. We compare this approach to other recently developed enhanced thixotropic Oldroyd-8 inspired models.

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I. INTRODUCTION

Many fluids and complex materials possess unique properties that render typical rheological models unsuitable for effective analysis. These particular fluids may demonstrate non-Newtonian properties by the nature of their viscosity and elasticity or may possess a microstructure that interacts with the solvent "*backbone*" to alter fluid behavior. One example of such a fluid is human blood due to the properties and interactions between its component red blood cells, white blood cells, plasma, and other materials. Of these components, red blood cells constitute most of the blood and interact with each other, under certain conditions, to form microstructures suspended in blood plasma called rouleaux. Recent work modeling the rheological behavior of human blood indicates that blood has all of the hallmark features of a complex material, including shear-thinning, viscoelastic behavior, yield stress, and thixotropy.[1] This anomalous behavior can be attributed to the formation, breakdown, and evolution of rouleaux structures, cylindrical, "*coin-like*" stacks of red blood cells. These typically form amid the re-structuring of blood at low shear. At low shear, fibrinogen proteins may act to link the faces of red blood cells to one another to form an aggregation reminiscent of a coin stack.[1,2] The formation and breakdown of rouleaux is a function of stress and shear rate, further indicating that blood possesses thixotropic properties. Blood is best described as a thixo-elasto-visco-plastic fluid (TEVP).[5]

The status of blood as a TEVP has important implications in its observed rheological behavior and the methods by which it is analyzed. A notable behavior of some TEVP fluids such as blood is a predilection toward shear thinning, a phenomenon by which the viscosity of a fluid decreases under greater applied shear stresses.[9,22,23] In blood, this is due to the breakdown of rouleaux microstructures, which allows the fluid to yield, and flow more freely under flow.[20] Understanding the variable and changing nature of this blood property is vital in fully understanding the nature of its complexity. A clear example of this phenomenon lies in the body's clotting process, where the viscosity of blood increases greatly in proximity to a wound, eventually forming a solid plug at the point of breach. In addition to hematocrit, fibrinogen, and cholesterol levels, various

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health conditions and pharmaceuticals have been shown to affect the shearing properties of blood, which may subsequently induce the body's vascular system to modulate flow.[1,3,5,19,22] Over long periods, the increased stress on the vascular system can lead to cardiovascular issues.[3]

The status of blood as a TEVP fluid additionally entails that has a yield stress. As such, the component cells of blood will not deform without a measure of applied stress,[4] whereas Newtonian fluids would tend to deform without the application of a minimum degree of preserving force. In light of the TEVP properties of blood, it is understood that the body's cardiovascular system must actively apply work to induce in blood the yield necessary for circulation.[20] Relevant to the experimental tasks discussed in this paper, this phenomenon of work-induced deformation must be replicated in any *ex vivo* experimentation utilizing a rheometer to actively study the properties of blood.[5]

In addition to the inherent fluid properties of blood, the various microstructures present within the blood contribute to the total stress of the blood. This hails to the "thixo-" in TEVP—the feature of thixotropy—which describes the evolving nature of the microstructure in the formation of rouleaux at low shear and its subsequent breakdown under low shear conditions.[6,7] The "elasto-" refers to blood's elastic properties before deformation.[7,8] The "visco-" refers to the viscosity of blood, which increases its resistance to free flow, offsetting the effects of stress for a short time. Finally, the "plastic" refers to the plasticity of blood's component rouleaux, which deform and, sometimes, undergo irreversible deformation.[9,22]

Classical steady-state modeling of blood has enjoyed extensive development throughout the latter century, as demonstrated by models ranging from the Carreau—Yasuda method to the Herschel—Bulkley model. However, while these models certainly possess a range of assumptions to assist in the elimination of certain variables, recent work has given rise to a newer set of models better suited to characterizing the evolving nature of both transient and steady-state flow within TEVP materials.[1,7,9,19] This new generation of models frames a set of codependent differential equations within a given timeframe, accompanied by algebraic constraints, to describe several properties of the TEVP matter. In the development of a new model, the established Horner–Armstrong–Wagner–Beris (HAWB and mHAWB)[5] model is used as a basis upon which modifications are made to the subject Giesekus model. Specifically, the thixotropic features of the mHAWB model, primarily used to model the elastic and viscoelastic effects of the evolving microstructure, are applied to the Giesekus model framework.[10,11]

While the base Giesekus model equations act to model the viscoelastic properties of fluid, and a complex material under applied stress, the elastic and viscoelastic stress contributions from the blood microstructure such as rouleaux are then linearly superimposed. This contribution from rouleaux is added to the Giesekus solution to produce a more robust representation of the rheological properties of blood.[12] Subsequently, an analysis is performed to gauge the model's ability to predict unidirectional and small/large amplitude oscillatory shear flow. The experimental portion is then followed by the discussion of some rheological methods by which models could be fit to blood's non-linear behavior in varying shear rates.[10,13–15] The incorporation of the scalar structure evolution parameters is detailed, with subsequent comparison of the fit results of the new transient blood data to that of the model.[9,19] Particular emphasis is placed on the model's fitting procedure, first fitting to the constant steady state before methodically fitting the remaining transient parameters via a series of 4× step-ups and 4× step-downs in shear rate experiments. All parameters having been optimized via a global stochastic parallel tempering algorithm, the LAOS and UD-LAOS and oscillatory shear flow predictions are compared to data. The efficacy of the modified Giesekus model is then compared to other models as well as a critical assessment of the model's parameterization.[5,13,17,20,21]

The Giesekus model employed in this effort is a further development of the previous linear Maxwell model by incorporating additional terms.[16] The linear Maxwell model, in turn, is a derivative of a spring and dashpot model, which features elastic and viscous behavior for the respective components upon application of stress to the system.[8] Furthermore, the linear Maxwell model includes a tensor form to account for the three-dimensional aspect of actual stress-induced strain.[18] While the linear Maxwell, in itself, is quite a simple system, there exist other analogous, more complex systems for fluid modeling based on the spring-dashpot concept.[8] However, many of these systems do not add a large degree of precision to the modeling of TEVP fluids such as blood despite their added complexity and parameter cost. The linear Maxwell provides a sufficient basis among the "simpler" models for thixotropic enhancement. These modifications would help address the model's current inability to provide sufficient modeling capabilities for blood transient shear behavior.[1] The added nonlinear stress term of the Giesekus model enhances it for the nonlinear variations in stress that occur in fluids such as blood. However, the equation of every descendant model of the linear Maxwell model, including Giesekus, possesses a variation of its parent's three terms. To this, we add our thixotropic features of which the first term represents shear breakage, which could represent phenomena such as the dissolution of microstructures such as rouleaux. The second describes shear aggregation, which describes the interactions of nearby microstructures, given the shear rate. The third component of these thixotropic equations is a result of the Brownian aggregation, where red blood cells under the influence of random Brownian motion spontaneously aggregate.[5,9,13]

We compare results to the mHAWB model, a derivative of the established HAWB model.[5,19] The mHAWB model utilizes the tensorial response elements of the White–Metzner along with the cross-equation to better model the transient state behavior of blood.[20] The original HAWB assumed codependence between the elastic and viscoelastic responses and used 11 parameters to facilitate calculation, whereas the mHAWB model dispenses with a single structure parameter while adding a unique time dependence for the viscoelastic response.[5] The original Giesekus model and the enhanced thixotropy framework proposed here make use of 11 distinct parameters, along with algebraic relationships to determine material parameters.[11] This number of parameters exceeds that of many of its enhanced thixotropic contemporaries in the Maxwell family such as the correlational Jefferies.[11,21]

The proposed enhanced thixotropic Giesekus model could act to further enhance the model's capability for solving components of the full stress tensor, allowing it to better model TEVP-type fluids. The thixotropic framework allows the Giesekus model to account for the elastic and viscoelastic factors of stress due from microstructures as it evolves within the blood. The new model will be compared

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with the enhanced thixotropic correlational Jefferies, mHAWB, convected Maxwell, and Olroyd-4 models to determine the Giesekus model's efficacy.[21]

The model details will be provided in Sec. II. Experimental protocols and fitting procedures will also be elaborated in Sec. III. Via the application of LAOS and UDLAOS to the acquired blood samples with a subsequent comparison to the models' output predictions suggests that while not as capable as mHAWB, the modified Giesekus model provided a stronger correlation to actual blood behavior than other Maxwell variants, albeit with the cost of additional parameters, for certain rheological flows. The relevant data will be further discussed in Sec. IV.

II. MODEL BACKGROUND AND FRAMEWORK

The Oldroyd 8-constant (O8) framework[2,11,20,21] is used to compute the viscoelastic backbone contribution to total stress from the plasma of human blood. This quantity is then linearly superimposed with both the elastic and viscoelastic contribution to total stress from the rouleaux, or microstructure of human blood. The paradigm is given by (Eq. 8 of Ref. 21 and Eq. 7.3-2 of Ref. 20)

$$\begin{aligned}\boldsymbol{\tau} + \lambda_1\boldsymbol{\tau}_{(1)} + \frac{1}{2}\lambda_3\left\{\boldsymbol{\gamma}_{(1)}\cdot\boldsymbol{\tau} + \boldsymbol{\tau}\cdot\boldsymbol{\gamma}_{(1)}\right\} \\ + \frac{1}{2}\lambda_5(\mathrm{tr}\,\boldsymbol{\tau})\boldsymbol{\gamma}_{(1)} + \frac{1}{2}\lambda_6\left(\boldsymbol{\tau}:\boldsymbol{\gamma}_{(1)}\right)\boldsymbol{\delta} \\ = -\eta_0\left[\boldsymbol{\gamma}_{(1)} + \lambda_2\boldsymbol{\gamma}_{(2)} + \lambda_4\left\{\boldsymbol{\gamma}_{(1)}\cdot\boldsymbol{\gamma}_{(1)}\right\}\right. \\ \left. + \frac{1}{2}\lambda_7\left(\boldsymbol{\gamma}_{(1)}:\boldsymbol{\gamma}_{(1)}\right)\boldsymbol{\delta}\right],\end{aligned} \tag{1}$$

where boldface indicates a tensor, (1) and (2) indicates the *first* and *second* lower convected time derivative of stress and strain, tr is the trace, a single or double dot is the dot or double dot product, respectively, and $\boldsymbol{\delta}$ is the Kronecker delta. There are eight constants in the O8 framework: $\eta_0, \lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5, \lambda_6$, and λ_7.[2,20,11,21] Saengow and coworkers use the following paradigm for the O8 constants: $\eta_0, \lambda_1, \lambda_2, \mu_1, \mu_2, \mu_0, \nu_1$, and ν_2.[2,20,11,21] As a basis of comparison, all results will be compared to the previous results for the same donors in Ref. 21. According to Bird *et al.*,[20] there were no rheological reasons not to include nonlinear terms in stress in constitutive equations, and here we introduce the nonlinear Giesekus as follows:

$$\tau_{tot} = \tau_s + \tau_p, \tag{2}$$

$$\tau_s = \eta_s\dot{\gamma}, \tag{3}$$

and

$$\tau_p + \lambda_1\dot{\tau}_p - \alpha\frac{\lambda_1}{\eta_p}\{\tau_p\cdot\tau_p\} = \eta_p\dot{\gamma}, \tag{4}$$

where τ_s and τ_p are the contribution to total stress from the plasma and contribution to stress from the single red blood cells, proteins, platelets (before forming rouleaux), etc., respectively, while η_s and η_p are the viscosities of each. λ_1 and α are the relaxation time and the "*nonlinear interaction*" parameter, respectively. As is done in Ref. 20, we combine Eqs. (2)–(4) by replacing τ_p in the last equation with $\sigma_{tot} - \sigma_s = \sigma_{tot} + \eta_s\dot{\gamma}$ leaving the following:

$$\begin{aligned}\boldsymbol{\tau} + \lambda_1\boldsymbol{\tau}_{(1)} - a\frac{\lambda_1}{\eta_0}\{\boldsymbol{\tau}\cdot\boldsymbol{\tau}\} - a\lambda_2\left\{\boldsymbol{\gamma}_{(1)}\cdot\boldsymbol{\tau} + \boldsymbol{\tau}\cdot\boldsymbol{\gamma}_{(1)}\right\} \\ = \eta_0\left[\boldsymbol{\gamma}_{(1)} + \lambda_2\boldsymbol{\gamma}_{(2)} - a\frac{\lambda_2^2}{\lambda_1}\left\{\boldsymbol{\gamma}_{(1)}\cdot\boldsymbol{\gamma}_{(1)}\right\}\right],\end{aligned} \tag{5}$$

where boldface represents the full tensor and the parentheses represent the convected time derivatives of the stress tensor, and $\gamma_{(1)}$ is the rate of strain tensor (λ_1 and a are defined in Table I).[20] We summarize the generic equations for the Giesekus model in Table I.

The Giesekus model is used to model the viscoelastic backbone of the plasma, individual red blood cells, and other constituents of blood. Table I describes the four ordinary differential equations for the three cardinal directions and the *yx* (direction of shear) for the Giesekus model. To adequately model the "coin-like" stacks of red

TABLE I. Viscoelastic equations for *xx*, *yy*, *zz*, and *yx* components.[2,20,21]

Par.	$\eta_0, \lambda_1, \lambda_2, \alpha$	
SS	$\tau_{ve,plasma} = \eta(\dot{\gamma})\dot{\gamma}$	$f = \frac{(1-\chi)}{(1+(1-2\alpha)\chi)}$
	$\frac{\eta}{\eta_0} = \frac{\lambda_1}{\lambda_2} + \left(1 - \frac{\lambda_2}{\lambda_1}\right)\left(\frac{(1-f)}{1+(1-2\alpha)f}\right)$	$\chi^2 = \frac{\left(1+16\alpha(1-\alpha)(\lambda_1\dot{\gamma})^2\right)^{1/2}-1}{8\alpha(1-\alpha)(\lambda_1\dot{\gamma})^2}$
Transient relationships	$\eta_0 = \eta_s + \eta_p$	$\lambda_2 = \lambda_1\frac{\eta_s}{\eta_s+\eta_p}$ $\quad a = \alpha\frac{(\eta_s+\eta_p)}{\eta_p}$
xx	$\frac{d\tau_{xx}}{dt} = \frac{1}{\lambda_1}\left[-\eta_0\left(2\lambda_2 + \frac{a\lambda_2^2}{\lambda_1}\right)\dot{\gamma}^2 + (2\lambda_1 + 2a\lambda_2)\tau_{yx}\dot{\gamma} + \frac{a\lambda_1}{\eta_0}\left(\tau_{xx}^2 + \tau_{yx}^2\right) - \tau_{xx}\right]$	
yy	$\frac{d\tau_{yy}}{dt} = \frac{1}{\lambda_1}\left[\eta_0\lambda_4\dot{\gamma}^2 + 2a\lambda_2\tau_{yx}\dot{\gamma} + \frac{a\lambda_1}{\eta_0}\left(\tau_{yx}^2 + \tau_{yy}^2\right) - \tau_{yy}\right]$	
zz	$\frac{d\tau_{zz}}{dt} = \frac{1}{\lambda_1}\left[\frac{a\lambda_1}{\eta_0}\tau_{zz}^2\tau_{zz}\right]$	
yx	$\frac{d\tau_{yx}}{dt} = \frac{1}{\lambda_1}\left[-\eta_0(\dot{\gamma} + \lambda_2\ddot{\gamma}) - \frac{1}{2}\lambda_3\tau_{xx}\dot{\gamma} + \frac{1}{2}(2\lambda_1 + \lambda_3)\tau_{yy}\dot{\gamma} - \frac{a\lambda_1}{\eta_0}(\tau_{xx} + \tau_{yx}) - \tau_{yx} - \tau_{yx}\right]$	

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blood cells, called rouleaux, we will incorporate, as in our previous work,[18,19,21] a structure portion of the model that will adequately capture the continual rouleaux formation, breakdown, and their contribution to total stress. Here, we will use a linear superposition of the total viscoelastic stress from the plasma with individual red blood cells, given by the Giesekus model, and a viscoelastic and elastic contribution to total stress from the rouleaux as

$$\tau_{tot} = \tau_{yx,ve} + \tau_{r,ve} + \tau_{r,e}, \tag{6}$$

where $\tau_{yx,ve}$ is the viscoelastic contribution to total stress from the plasma and individual constituents, $\tau_{r,ve}$ is the viscoelastic contribution from rouleaux, and $\tau_{r,e}$ is the elastic contribution from the rouleaux.[5,18,19,21,22] For the last two terms in Eq. (6), we will incorporate a thixotropic model with a scalar thixotropic parameter, β, to communicate the level of structure present at any time, spanning a range of [0 1]. Where zero is no structure (or close to no structure) or no rouleaux, a value of 1 represents fully structured or full rouleaux development. Here, we follow the previous approach,[21] inspired by Horner and coworkers, to compute the elastic and viscoelastic stress contributions from the rouleaux.[5,19,21,22] Here, we present a streamlined set of the thixotropic equations; for more details, refer to Refs. 5, 19, 21, and 22.

The following equation describes the evolution of the structure as a function of the plastic shear rate of blood:

$$\frac{d\beta}{dt} = \frac{1}{t_\beta}\left(-t_{r1}\beta|\dot{\gamma}_p| + t_{r2}(1-\beta)|\dot{\gamma}_p|^{1/2} + (1-\beta)\right), \tag{7}$$

where t_{r1} and t_{r2} are the ratios of rouleaux shear breakdown and shear buildup, respectively, and t_β is the overall rouleaux rebuild constant.[5,9,19,21,22] Equation (7) has three terms to describe the evolution of thixotropy: (1) Shear breakdown, proportional to shear rate, and β, amount of the structure still present; (2) shear buildup, proportional to $(1-\beta)$; and (3) Brownian buildup, also proportional to $(1-\beta)$.[5,21] The advantage of the kinetic expression used in this article is that it is able to capture three important mechanisms of rouleaux breakdown and formation, which are as follows: (1) shear breakdown, which is a function of current structure and shear rate; (2) structure buildup due to shear term, which is a function of shear rate raised to the power of (1/2) and the amount of lone unstructured red blood cells; and (3) Brownian buildup to account for rouleaux formation that is a function of lone, unstructured red blood cells. This kinetic expression explicitly accounts for the physics of rouleaux breakdown and formation.[5,9,19,21,22] For more information with respect to the steady-state fitting procedure, see Refs. 5, 9, 19, 21, and 22.

This model incorporates the separation of strain and shear rate into elastic and plastic components from Dimitriou and co-workers, where $\gamma = \gamma_e + \gamma_p$ and $\dot{\gamma} = \dot{\gamma}_e + \dot{\gamma}_p$ and is further described by

$$\dot{\gamma}_p = \begin{cases} \dfrac{\dot{\gamma}}{\left(2 - \frac{\gamma_e}{\gamma_{max}}\right)}, \dot{\gamma} \geq 0 \\ \dfrac{\dot{\gamma}}{\left(2 + \frac{\gamma_e}{\gamma_{max}}\right)}, \dot{\gamma} < 0, \end{cases} \tag{8}$$

where a subscript e represents the elastic component and a subscript p represents the plastic component.[5,9,19,21,22] The plastic components are involved in irreversible processes and breakage of structure, while the elastic components are involved in reversible processes.[5,9,19,21,22] The maximum rouleaux strain is given as

$$\gamma_{max} = \gamma_0\beta, \tag{9}$$

where γ_0 is the rouleaux zero shear rate limiting elastic strain.[5,9,19,21,22] The elastic strain rate is given as

$$\dot{\gamma}_e = \begin{cases} \dot{\gamma}_p - \dfrac{\gamma_e}{\gamma_{max}}|\dot{\gamma}_p|, & \dfrac{d\gamma_{max}}{dt} \geq 0 \\ \dot{\gamma}_p - \dfrac{\gamma_e}{\gamma_{max}}|\dot{\gamma}_p| + \dfrac{\gamma_e}{\gamma_{max}}\dfrac{d\gamma_{max}}{dt}, & \dfrac{d\gamma_{max}}{dt} < 0. \end{cases} \tag{10}$$

The elastic contribution from rouleaux to total stress is

$$\tau_{r,e} = \frac{\tau_{y,0}}{\gamma_0}\beta, \tag{11}$$

where τ_{y0} is the yield stress.[5,9,19,21,22] The viscoelastic contribution from the rouleaux to the total stress is now defined as follows:

$$\frac{d\tau_{r,ve}}{dt} = \begin{cases} G_r\beta^m\left(\dot{\gamma}_p - \dfrac{\tau_{r,ve}}{\mu_r\beta^m}\right), & \dfrac{d\beta}{dt} \geq 0 \\ G_r\beta^m\left(\dot{\gamma}_p - \dfrac{\tau_{r,ve}}{\mu_r\beta^m}\right) + m\dfrac{\tau_{r,ve}}{\beta}\dfrac{d\beta}{dt}, & \dfrac{d\beta}{dt} < 0, \end{cases} \tag{12}$$

where G_r is the elastic modulus of the rouleaux, μ_r is the rouleaux structural viscosity, and m is a fitting parameter (in the literature usually close to 1.5).[5,9,19,21,22] The steady-state value of the structure parameter and the steady-state stress are defined in the following equations:

$$\beta_{ss} = \frac{\left(t_{r2}|\dot{\gamma}|^{1/2} + 1\right)}{\left(t_{r1}|\dot{\gamma}| + t_{r2}|\dot{\gamma}|^{1/2} + 1\right)}, \tag{13}$$

$$\tau_{ss} = \tau_{ve,ss} + \mu_r\beta_{ss}^m + \tau_{y,0}\beta_{ss}. \tag{14}$$

This completes the description of the elastic and viscoelastic contributions from the rouleaux to the total stress.[5,9,19,21,22] We point out here that at steady state, the rouleaux are no longer evolving and, therefore, use the actual shear rate. For a more detailed explanation of rouleaux equilibrium at steady state, see Refs. 5, 9, 19, 21, and 22.

III. EXPERIMENTAL METHODS

A. Materials and sample preparation

The handling and rheological measurement protocol employed for the blood rheology in this work follow previously established guidelines for blood rheology and that described by Horner and co-workers.[5,19,21,22] This work is in compliance with the Institutional Review Board of the University of Delaware.[5,19,21,22] A complete and detailed description of the blood draw procedure can be found in Refs. 5, 19, 21, and 22. Lab results are given in Table II.[5,19,21,22]

All measurements were performed with an ARES-G2 strain-controlled rheometer from TA Instruments equipped with a double wall couette.[5,9,19,21,22,32] The dimensions and measurable range

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TABLE II. Physiological parameter values of five donors.

Donor	Hematocrit (%)	Fibrinogen (mg/dl)	Total chol. (mg/ml)	Triglycerides (mg/ml)	HDL chol. (mg/ml)	LDL chol. (mg/ml)
1	42.9	0.242	138	108	35	83
2	46.2	0.316	132	111	41	72
3	38.3	0.333	145	144	39	82
4	45.2	0.214	187	57	65	108
5	43.1	0.249	244	94	70	154

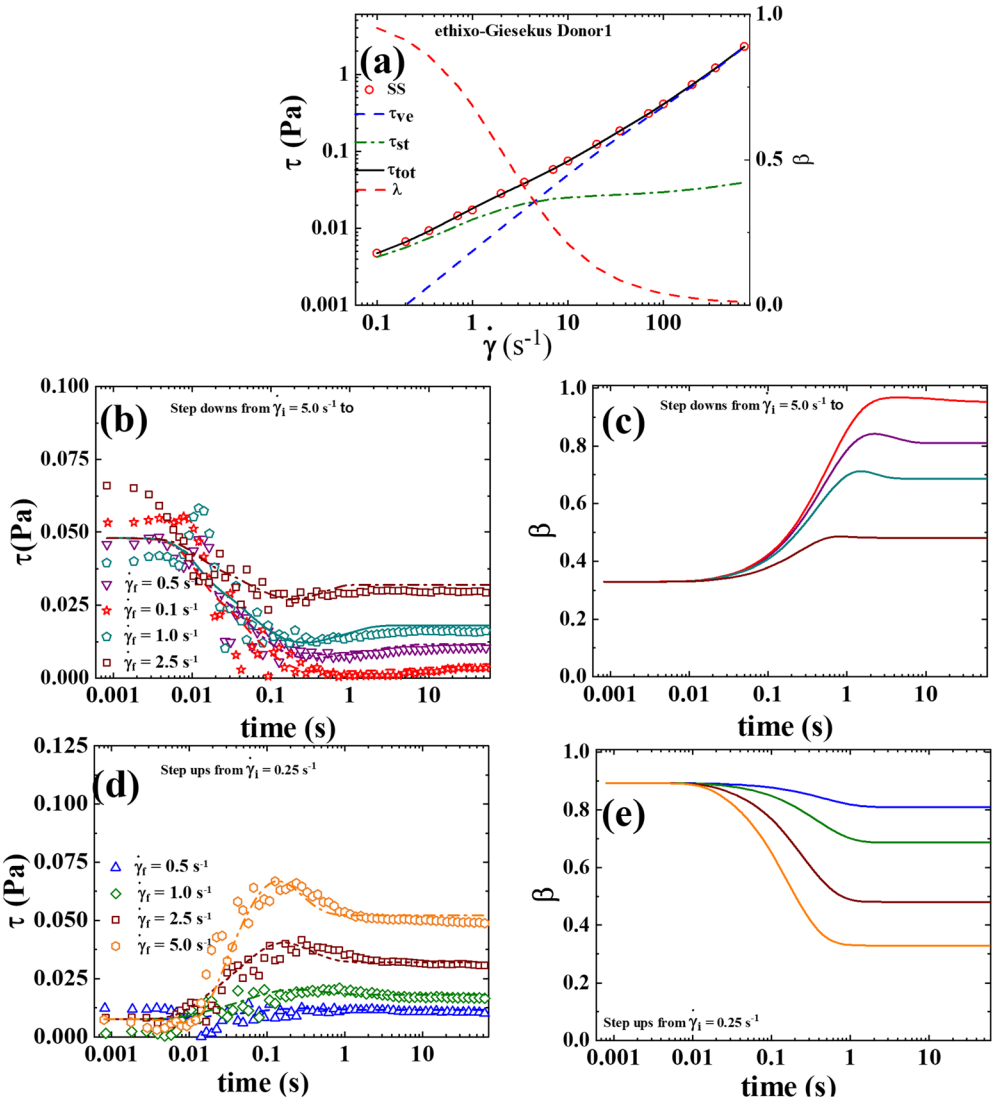


FIG. 1. ethixo-Giesekus model fit for (a) steady-state human blood data; (b) set of 4 step-downs in the shear rate from $\dot{\gamma} = 5.0$ to 0.1, 0.5, 1.0 and 2.5 s^{-1}; (c) corresponding structure parameter curves (colors of structure parameter curves match with respective colors from stress evolution curves); (d) step-ups in the shear rate from $\dot{\gamma} = 0.25$ to 0.5, 1, 2.5, and 5 s^{-1}; and (e) corresponding structure parameter curves (donor1) (dataset1).[24]

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can be found in Ref. 5, 9, 19, 21, and 22. The specifications and details of the rheological protocol can be found in Refs. 5, 9, 19, 21, and 22. As in our previous work, we again show experimental results for steady state, step-up/step-down in shear rate tests, small amplitude oscillatory shear (SAOS), large amplitude oscillatory shear (LAOS), and uni-directional large amplitude oscillatory shear flow (UD-LAOS).[5,9,19,21,22,32] Our experimental procedures and protocols follow exactly the work of Horner and co-workers, and all of the rheological data for the five donors in Table I can be found on Mendeley.[5,9,19,21,22,32] The LAOS and UD-LAOS experiments presented in this work follow the same protocol as described by Horner and co-workers[5,9,19,21,22,32] and Armstrong *et al.*,[9] where the strain and shear rates are defined as

$$\gamma(t) = \gamma_0 \sin(\omega t) \tag{15}$$

and

$$\dot{\gamma}(t) = \gamma_0 \omega \cos(\omega t), \tag{16}$$

for the LAOS experiments and

$$\gamma(t) = \gamma_0 \sin(\omega t) + t\gamma_0 \omega, \tag{17}$$

$$\dot{\gamma}(t) = \gamma_0 \omega \cos(\omega t) + \gamma_0 \omega, \tag{18}$$

for the UD-LAOS experiments. The UD-LAOS data and model fits will be plotted using effective Lissajous–Bowditch elastic and viscous projections, where the measured stress is a function of the oscillatory component of the strain and shear rate as shown in Eqs. (17) and (18), respectively.[5,9,19,21,22,32]

The small amplitude oscillatory shear (SAOS) follows the standard formalism

$$\tau_{yx}(t) = \gamma_0 \sum_{i,odd}^{n} \left(G_i' \sin(n\omega t) + G_i'' \cos(n\omega t) \right), \tag{19}$$

where $n = 1$, for SAOS, and Eq. (19) is a function of time t, frequency ω, and strain amplitude γ_0. In addition, it is noted that, for LAOS, n is the running index (when not equal to 1), and the upper limit of the summation goes to infinity. We use the ratio of third to first harmonic as a metric of success of our model predictions as

$$\frac{I_3}{I_1} = \sqrt{\frac{(G_3'^2 + G_3''^2)}{(G_1'^2 + G_1''^2)}}, \tag{20}$$

where the subscript 3 is the third harmonic value.[5,19] Equation (20) is the "lumped measure," known as the relative intensity of the third harmonic normalized by the first harmonic intensity and will be used as a metric of model efficacy, and shown on the SAOS predictions for each model in Figs. 2(a) and 2(b).[5,9,19,21,22,32]

B. Parameter optimization

The enhanced thixotropic Giesekus will incorporate identical techniques as found in Refs. 18, 19, and 21. We present a streamlined explanation here. For the model fitting procedure, we follow the previous protocol as established by Armstrong and Tussing, originally inspired by Horner and co-workers with the following exceptions:[5,19,21,22] (1) we fix the steady-state parameters to steady-state data by minimizing the *normalized* cost function, $F_{cost,ss}$, using a parallel tempering algorithm, as follows:[21,24,28]

$$F_{cost,ss} = \frac{1}{N} \sqrt{\sum_{i=1}^{N} \left(\frac{(y_i - f_i)}{y_i} \right)^2}, \tag{21}$$

where y_i is the steady-state stress data and f_i is the model prediction; (2) while keeping the steady-state parameters fixed, then fit the transient parameters to *four* sets of step-up in shear rate and *four* sets of step-down in shear rate experiments, again with parallel tempering, minimizing the following cost function:

$$F_{cost,trans} = \frac{1}{M} \sum_{k=1}^{M} \frac{1}{N} \sqrt{\sum_{i=1}^{N} (y_i - f_i)^2}, \tag{22}$$

where M is the total number of transient step-ups/step-downs in shear rate tests and N is the number of points per experiment.[21] A more rigorous explanation of the fitting procedure is provided in Ref. 21. The predictions for each set of rheological data are shown in Fig. 2 (and original figures in Ref. 21), with cost functions computed

TABLE III. Best fit parameters and F_{cost}, RSS, AIC, and BIC for donor1 with ethixo-Giesekus.

Par.	Donor1
η_0 (Pa s)	0.0051
η_s (Pa s)	0.0031
η_p (Pa s)	0.0020
λ_1 (s)	0.0270
λ_2 (s)	0.0164
Alpha	0.3068
tr_1 (s)	0.5081
tr_2 ($s^{1/2}$)	0.1126
μ_R (Pa s)	0.0181
τ_{y0} (Pa)	0.0027
m (–)	1.2812
τ_β (s)	0.61
G_R(Pa)	0.25
γ_∞	1
F_{cost} (SS)	0.0037
n (points)	16
RSS	0.004
k (par.)	11
AIC	10.72
BIC	49.72
F_{cost} (trans)	0.00056
n (points)	390
RSS	0.047
k (par.)	13
AIC	19.89
BIC	149.01

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for each set of LAOS and UD-LAOS as

$$F_{cost,LAOS} = \frac{1}{N}\sqrt{\sum_{i=1}^{N}\left(y_i - f_i\right)^2}. \quad (23)$$

The small amplitude oscillatory shear is predicted, and $F_{cost,\,SAOS}$ is computed as follows:

$$F_{cost,SAOS} = \sum_{i=1}^{n}\left(\sqrt{(G'_{i,data} - G'_{i,model})^2} + \sqrt{(G''_{i,data} - G''_{i,model})^2}\right)/(2n), \quad (24)$$

where n is the number of points. $F_{cost,\,SAOS}$ is calculated for both amplitude and frequency sweeps. The fit model parameters are held constant and used to predict a full period at alternance, and using $\boldsymbol{A} = \boldsymbol{bx}$, where it is an array consisting of two columns: $\gamma_0 \sin(\omega t)$ and $\gamma_0 \cos(\omega t)$ and $\boldsymbol{b}$ is the stress prediction over a period.[21] In addition, the third harmonic moduli are calculated (G'_3 and G''_3) and are shown in Fig. 2.

Each of the LAOS and UD-LAOS predictions accumulate $F_{cost,\,LAOS}$ over three periods of data, with the Lissajous–Bowditch elastic and viscous projections showing the last period data and predictions. For the analysis, we incorporate the Akaike Information Criteria (AIC) and the Bayes Information Criteria (BIC) (with the steady-state and transient step fits) to allow for a way to balance additional parameters with accuracy.[21,29–31] The AIC and BIC penalize the models for additional parameters, allowing for a more objective comparison. The residual sum of squares, AIC, and BIC are computed as

$$RSS = \sum_{i=1}^{N}\sqrt{\left(y_i - f_i\right)^2}, \quad (25)$$

where y_i is the data and f_i is the model prediction. We note here again that the steady-state cost function and residual sum of squares are normalized by the data value,[21,29–31]

$$AIC = 2k + 2\ln(RSS), \quad (26)$$

$$BIC = 2k\ln(n) + 2\ln(RSS), \quad (27)$$

where RSS is the residual sum of squares, k is the number of parameters, and n is the number of data points.[21,29–31]

TABLE IV. Best fit ethixo-Giesekus parameters for all donors (1–5).[21]

Par.	1	2	3	4	5	AVE.	STDEV
η_0 (Pa s)	0.0051	0.0080	0.0047	0.0048	0.0075	0.0060	0.0016
η_s (Pa s)	0.0031	0.0036	0.0030	0.0035	0.0034	0.003	0.000
η_p (Pa s)	0.0020	0.0044	0.0018	0.0013	0.0041	0.003	0.001
λ_1 (s)	0.0270	0.2133	0.3982	0.0187	0.2956	0.191	0.167
λ_2 (s)	0.0164	0.0955	0.2491	0.0135	0.1339	0.102	0.097
Alpha	0.3068	0.0062	0.0005	0.1922	0.0057	0.102	0.140
tr_1 (s)	0.5081	0.3063	0.4531	0.3016	0.5618	0.426	0.118
tr_2 ($s^{1/2}$)	0.1126	0.0240	0.2038	0.1351	0.0035	0.096	0.082
μ_R (Pa s)	0.0181	0.0275	0.0217	0.0179	0.0487	0.027	0.013
τ_{y0} (Pa)	0.0027	0.0022	0.0021	0.0030	0.0037	0.003	0.001
m (−)	1.2812	1.9355	1.8814	1.4506	1.6370	1.637	0.278
τ_β (s)	0.61	1.13	1.67	0.65	0.5921	0.931	0.467
G_R(Pa)	0.25	0.77	0.43	0.28	0.5362	0.452	0.211
γ_∞	1	1	1	1	1	1	
F_{cost} (SS)	0.0037	0.0187	0.0052	0.0109	0.0221	0.0121	
n (points)	16	16	16	16	16		
RSS	0.004	0.090	0.007	0.031	0.12	0.051	
k (par.)	11	11	11	11	11		
AIC	10.72	17.17	12.06	15.02	17.84	14.57	
BIC	49.72	56.17	51.06	54.02	56.84	53.56	
F_{cost} (trans)	0.00056	0.00072	0.00051	0.00048	0.00122	0.00070	
n (points)	390	390	390	390	390		
RSS	0.047	0.079	0.040	0.035	0.23	0.0853	
k (par.)	13	13	13	13	13.00		
AIC	19.89	20.91	19.54	19.28	23.03	20.53	
BIC	149.01	150.03	148.66	148.40	152.15	149.65	

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IV. RESULTS AND DISCUSSION

The enhanced thixotropic Giesekus has a total of 11 parameter fits to the steady state, and three more fits to a total of eight transient step-ups/step-downs in shear rate experiments, with the last parameter $\gamma_{0,R}$ being set to 1 as recommended in the literature.[21,24,28] Again, we follow the analysis found in Ref. 21. Figure 1(a) shows the steady-state fit, and Figs. 1(b) and 1(d) show the fit to the four step-ups and four step-downs in shear rate experiments, respectively. Figures 1(c) and 1(e) show the corresponding structure parameter evolution during the step-up and step-down in shear rate experiments.[21,24,28] Both Figs. 1(b) and 1(c) show that qualitatively the ethixo-Geisekus can adequately capture the stress of the step-up and step-down experiments. Table III shows the best fit model parameters from the steady-state and transient data fits, as well as the value of F_{cost}, *RSS, AIC, and BIC* for donor 1. The value of the steady-state F_{cost} value is dimensionless because the difference between model prediction and data was normalized with the value of each stress measurement of each data point for optimized fitting.[21,24,28] The parameters, average and standard deviations of parameter fitting, and F_{cost}, *RSS, AIC, and BIC* are shown for all donors in Table IV. Figures 2(a)–2(d) show the model predictions to the SAOS, LAOS, and UDLAOS experiments.[21,24,28] The series of figures are depicting a set of Lissajous–Bowditch elastic and viscous projections. The LAOS and UDLAOS predictions, which are qualitatively seen in Figs. 2(a) and 2(b) and qualitatively noted in Table V, are also modest.[21,24,28]

Comparing Tables V and VI, it is interesting to note that the additional parameters of the enhanced thixotropic Giesekus do imbue a better value for average cost function of the steady-state fits as the average F_{cost} values indicate. With respect to the set of eight transient step-ups/step-downs in shear rate, the additional parameters do not decrease the average value of cost function. However, with respect to the oscillatory shear flow average F_{cost} values as shown in Tables V and VI, the ethixo-Giesekus is now performing ahead of its peer models, and this contributes to the F_{cost}, overall, of the ethixo-Giesekus to be the second-best model overall with the mHAWB slightly better. Acknowledged here is the fact that, in this paper, we fit the ethixo-Giesekus to five donors, but in the previous comparison work,[20] only four donors were incorporated. Regardless,

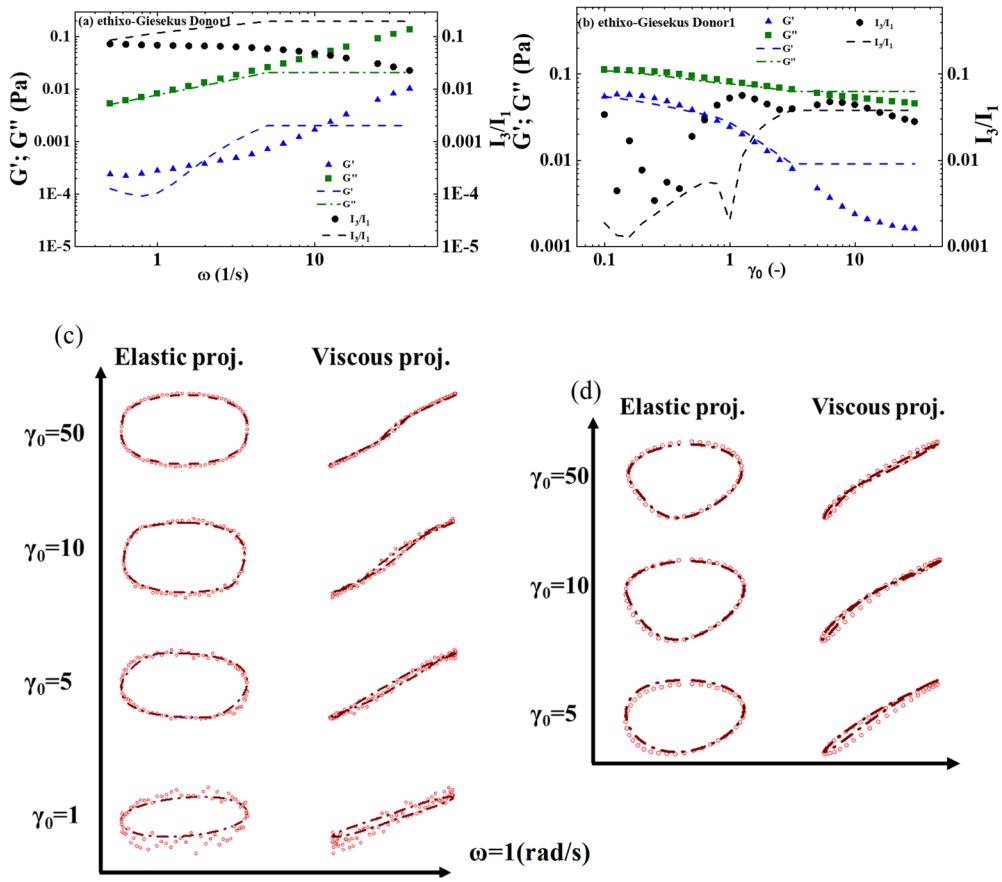


FIG. 2. (a) Frequency sweep at $\gamma_0 = 10(-)$ and (b) amplitude sweep at $\omega = 12.566$ (rad/s) (green and blue points: data and green and blue lines: ethixo-mHAWB predictions), (c) elastic projections and viscous LAOS projections of data (red dots) and ethixo-Giesekus predictions (solid green lines), and (d) effective elastic and viscous UDLAOS projections at $\omega = 1(\text{rad/s})$; $\gamma_0 = 5, 10,$ and $50(-)$ (donor1) (dataset1).[24]

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TABLE V. ethixo-Giesekus comparison of average F_{cost}, AIC, and BIC (average of ***five*** donors).

	ethixo-Giesekus
F_{cost} (SS)	0.0121
RSS	0.033
k (par.)	11
AIC	14.565
BIC	53.562
F_{cost} (trans)	0.00070
RSS	0.08527
k (par.)	13
AIC	20.530
BIC	149.649
$F_{cost, LAOS}$ (Pa)	0.00085
$F_{cost, UDLAOS}$ (Pa)	0.00085
$F_{cost, SAOS}$ (Pa)	0.00251
$F_{cost,tot}$ (Pa)	0.00123

the average values of F_{cost}, RSS, AIC, and BIC are used for comparison. This is tempered by the fact that there is a cost for the additional parameters as calculated by the AIC and BIC of the steady-state and transient parameter fits.[21,29,30]

In addition to the F_{cost} value comparison of the enhanced thixotropic Giesekus to previous Oldroyd-8 enhanced thixotropic models,[21] we combine best fit model parameters from Table IV and physiology parameters from Table II for each donor, to perform an parametric analysis via the correlation matrix. Table VII shows the calculation of the 95% confidence interval for each of the parameters. This could be used to determine, via "*bracketing*" normal, healthy model parameter values, if a donor potentially has a pathological condition. This could be potentially indicated by falling outside of the 95% confidence interval. One could easily manipulate the confidence interval level here to obtain results that could trigger a further medical evaluation of a donor with a medical condition that is impacting the behavior of the blood. We note that in Table VII, the value of alpha is less than zero because of too few donors to conduct this analysis; however, future iterations could mitigate this problem.

Table VIII shows the correlation matrix (a), the p-value matrix, and the legend to interpret the correlation matrix. We compute both the correlation and p-value matrix using *corrcoef* command in Matlab. In addition to this, we note that Matlab can also deconvolute the effect of the parameters on each other by controlling the other parameters with the *partialcorr* command. Since we had minimal donors, and, therefore, minimal, statistically meaningful correlations, we chose the first option. Our analysis here is slightly hindered by the small number of donors included; however, one could ostensibly increase the statistical significance of the correlations by including more donors in the future. A cursory analysis of Tables VIII and IX shows that blood's hematocrit (volume percent red blood cells) is strongly correlated with the solvent viscosity, and the low-density lipoproteins are strongly correlated with the yield stress. We reiterate that this analysis contains only 5 donors; however, this finding is consistent with the literature with respect to human blood, which is not surprising. Moving forward, one could start to increase the amount of meaningful information with respect to correlations by increasing the set of eligible donors and begin to draw stronger inference about how the physiological parameters of human blood affect the very dynamic thixo-elasto-visco-plastic behavior of human blood.

TABLE VI. Comparison values of F_{cost}, RSS, AIC, and BIC of the enhanced thixotropic Maxwell, Corotational Jeffries, Oldroyd-4, and mHAWB.[21] Reproduced with permission from M. J. Armstrong and J. Tussing, Phys. Fluids, **32**, 094111 (2020). Copyright 2020 AIP Publishing LLC.

	ethixo-			
	Conv. Maxwell	Corot. Jeffries	Oldroyd-4	mHAWB
F_{cost} (SS)	0.01601	0.00844	0.01097	0.00619
k (par.)	6	8	9	9
RSS	0.070	0.021	0.034	0.010
AIC	8.43	7.75	10.82	8.63
BIC	33.25	36.11	42.73	40.54
F_{cost} (trans.) (Pa)	0.00051	0.00052	0.00076	0.00052
k (par.)	10	11	12	12
RSS	0.0423	0.052	0.087	0.044
AIC	13.45	15.51	19.10	17.52
BIC	112.77	124.77	138.29	136.71
$F_{cost, LAOS}$ (Pa)	0.00106	0.00060	0.00133	0.00066
$F_{cost, UDLAOS}$ (Pa)	0.00119	0.00082	0.00243	0.00080
$F_{cost, SAOS}$ (Pa)	0.00309	0.00687	0.00298	0.00169
$F_{cost,tot}$ (Pa)	0.00146	0.00220	0.00187	0.00092

TABLE VII. Depiction of average and standard deviation of enhanced thixotropic Giesekus model parameters using 5 donors, with 95% confidence intervals ($\bar{x} + / - Z\frac{s}{\sqrt{n}}$), where *Z* is set to 1.959 96 for the 95% confidence, *s* is the standard deviation, and *n* is the number of donors.

	AVE	STDEV	−95%	95%
η_0 (Pa s)	0.0060	0.0016	0.0046	0.0074
η_s (Pa s)	0.0033	0.0003	0.0031	0.0035
η_p (Pa s)	0.0027	0.0014	0.0015	0.0040
λ_1 (s)	0.191	0.167	0.045	0.337
λ_2 (s)	0.102	0.097	0.016	0.187
Alpha	0.102	0.140	−0.021	0.225
tr_1 (s)	0.426	0.118	0.323	0.530
tr_2 ($s^{1/2}$)	0.096	0.082	0.024	0.168
μ_R (Pa s)	0.027	0.013	0.015	0.038
σ_{y0} (Pa)	0.003	0.001	0.002	0.003
m (−)	1.637	0.278	1.393	1.881
τ_λ (s)	0.931	0.467	0.522	1.340
G_R (Pa)	0.452	0.211	0.267	0.636

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TABLE VIII. Correlation matrix.

Corr. Mat.	η_0 (Pa s)	η_s (Pa s)	η_p (Pa s)	λ_1 (s)	λ_2 (s)	Alpha	tr_1 (s)	tr_2 ($s^{1/2}$)	μ_R (Pa s)	σ_{y0} (Pa)	m (−)	τ_λ (s)	G_R(Pa)
HCT	0.519	0.896	0.391	−0.618	−0.755	0.198	−0.542	−0.659	0.062	0.246	−0.167	−0.606	0.298
FIB	0.217	−0.243	0.309	0.768	0.787	−0.682	−0.082	0.164	−0.003	−0.743	0.870	0.910	0.617
CHOL	0.253	0.275	0.225	0.129	0.009	−0.226	0.413	−0.430	0.764	0.925	−0.176	−0.518	−0.037
TRI	0.000	−0.643	0.140	0.707	0.762	−0.406	0.341	0.285	−0.021	−0.627	0.547	0.781	0.289
HDL	0.211	0.512	0.131	−0.047	−0.153	−0.183	0.052	−0.384	0.581	0.844	−0.157	−0.529	−0.027
LDL	0.256	0.217	0.238	0.123	0.000	−0.186	0.500	−0.448	0.783	0.944	−0.223	−0.548	−0.061

TABLE IX. p-value matrix of best fit ethixo-Giesekus and physiological parameters.

P	η_0 (Pa s)	η_s (Pa s)	η_p (Pa s)	λ_1 (s)	λ_2 (s)	Alpha	tr_1 (s)	tr_2 ($s^{1/2}$)	μ_R (Pa s)	σ_{y0} (Pa)	m (−)	τ_λ (s)	G_R(Pa)
HCT	0.370	0.039	0.515	0.266	0.140	0.750	0.346	0.226	0.921	0.689	0.788	0.279	0.627
FIB	0.725	0.693	0.612	0.130	0.114	0.205	0.895	0.792	0.997	0.150	0.055	0.032	0.268
CHOL	0.681	0.655	0.716	0.837	0.989	0.715	0.489	0.470	0.133	0.024	0.777	0.372	0.953
TRI	0.999	0.241	0.823	0.182	0.135	0.497	0.575	0.642	0.973	0.258	0.340	0.119	0.638
HDL	0.734	0.378	0.834	0.940	0.806	0.768	0.934	0.524	0.304	0.072	0.801	0.359	0.965
LDL	0.677	0.726	0.699	0.844	1.000	0.765	0.391	0.449	0.117	0.016	0.719	0.339	0.923

TABLE X. Legend to read correlation matrix (p-values <0.05 are significant).[21,29,30]

Mag. of value of Corr. coefficient	Meaning
<0.35	Weak
0.35–0.67	Modest
0.68–0.89	Strong
>0.9	Very strong
Negative values = negative correlation	

V. CONCLUSIONS

We found that with additional parameters, the enhanced thixotropic Giesekus was not able to better capture qualitatively and quantitatively the thixotropic, elastic, and viscous features of human blood at steady state and during the step-up/step-down transient experiments, and seemed to perform disproportionately better predicting oscillatory shear flow than the comparator models[20] as shown by the comparison of Tables V and VI. This is not a surprising finding. Any gains in accuracy of fits and predictions, however, from the additional parameters of the enhanced thixotropic Giesekus are offset by the penalties assigned by the AIC and BIC. From the ethixo-Giesekus model fit parameter values and physiological values from the lab tests of donors 1–5, we were able to construct a master matrix. Using this matrix, we constructed a correlation and p-value matrix to analyze statistically meaningful relationships between the physiology, rheology, and rheological model parameters. One can see that hematocrit, as per the literature, has a strong correlation with the solvent viscosity. In other words, with an increasing concentration of red blood cells, the blood is more viscous. Finally, we used the ethixo-Giesekus model to construct the 95% confidence interval, acknowledging that with more donors, the intervals will become more meaningful with respect to "bracketing" or "book-ending" healthy or pathological human blood rheology. However, this represents an effort to construct an understanding of how "normal" or non-pathological healthy blood should behave with respect to the parameters of the enhanced thixotropic Giesekus model.

AUTHORS' CONTRIBUTIONS

All authors contributed equally to this work.

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DATA AVAILABILITY

The data that support the findings of this study are openly available in Mendeley.com at https://data.mendeley.com/datasets/d43r5ydpx8/1,[24] https://data.mendeley.com/datasets/jdzpx6hy6r/1,[25] https://data.mendeley.com/datasets/948ffnypjs/1,[26] https://data.mendeley.com/datasets/s8w6s6f68b/1,[27] and https://data.mendeley.com/datasets/kpjk7rctmw/1.[32]

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